DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW



EVALUATION SCHEME & SYLLABUS
For

B. Tech. 3rd Year

MECHANICAL AND INDUSTRIAL ENGINEERING

ON

CHOICE BASED CREDIT SYSTEM (CBCS)

[Effective from session 2018-19]

DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

EVALUATION SCHEME

MECHANICAL AND INDUSTRIAL ENGINEERING

V SEMESTER

Sl No.	Subject Code	Subject Name	Teaching Deptt.	L-T-P	Th/Lab Marks	Sessional		Total	Credit
						Test	Assig /Att		
1	RAS501	MANEGIRIAL ECONOMICS	Applied Science	3---0---0	70	20	10	100	3
2	RAS502/ RUC501	SOCIOLOGY /CYBER SECURITY	Applied Science	3---0---0	70	20	10	100	3
3	RME504	Design of Machine Elements		3---0---0	70	20	10	100	3
4	RMI501	Thermal and Hydraulic Machines		3---1---0	70	20	10	100	4
5	RMI502	Applied Probability and Statics		3---0---0	70	20	10	100	3
6	RME052, 055, RMI051, 052	Deptt. Elective -I		3---1---0	70	20	10	100	4
7	RME554	Design and Simulation Lab		3---1---0	70	20	10	100	4
8	RMI551	Thermal and Hydraulic Machines lab		0---0---2	50		50	100	1
9	RMI552	Applied Probability and Statics Lab		0---0---2	50		50	100	1
10	RMI559	Seminar						100	1
	TOTAL							**1000**	**24**

Department Elective Course –I

1. RMI051: ValueEngineering.
2. RMI052: Maintenance Engineering &Management
3. RME052: Mechtronics&Microprocessor.
4. RME055:Advance ManufacturingScience.

DESIGN OF MACHINE ELEMENTS

UNIT I

Introduction

Design requirements of machine elements, Principles of mechanical design, Modes of failures and Factor of Safety. Systematic design process, Aesthetic and Ergonomic considerations in design, Use of standards in design, Manufacturing consideration in design, Selection of preferred sizes, Indian Standards designation, Designation of carbon & alloy steels, Selection of materials for static and fatigueloads.

UNIT II

Design for Static and Fluctuating Loads

Cyclic stresses, Fatigue and endurance limit, Notch sensitivity, Stress concentration. Design for finite and infinite life, Soderberg, Goodman & Gerber criteria.

Shafts, Keys and Couplings

Materials for shaft, Stresses in shafts, Design of shafts subjected to twisting moment, bending moment and combined twisting and bending moments, Shafts subjected to fatigue loads. Types of keys, splines, Selection of keys, Strength of keys, Design of rigid and flexible couplings.

UNIT III

Design of Mechanical Drives

Introduction to power transmission and drives. Classification of gears, Terminology, Gear tooth proportions, Tooth forms, System of gear teeth, contact ratio, Standard proportions of gear systems, Interference in involute gears, Backlash, Selection of gear materials, Gear manufacturing methods, Failure of gear tooth, Design considerations, AGMA and Indian standards, Beam strength and wear strength of gear tooth, Design of spur and helical gears.

UNIT IV

Mechanical Springs

Classification of springs, Material for helical springs, End connections for compression and tension helical springs, Design of helical springs subjected to static and fatigue loading.

Power Screws

Forms of threads, multiple threads, Efficiency of square threads, Trapezoidal threads, Stresses in screws, Design of screwjack.

UNIT V

Basic of Finite Element Method, Variational calculus, Integral formulation, variational methods: Methods of weighted residuals, Approximate solution using variational method, Modified Galerkin method, Boundary conditions.

Basic Finite Element Concepts: Basic ideas in a finite element solution, General finite element solution procedure, Finite element equations using modified Galerkin method, Axis symmetric Problems, Axial bars, Torsion bars, Application in Solid Mechanic Problems, Plane truss problems using software application.

Note: Design data book is allowed in the examination

Text Books:

1. Design of Machine Elements,V.B. Bhandari, Tata McGraw HillCo.
2. Design of Machine Elements, Sharma and Purohit,PHI.
3. Machine Design, P C Gope, PHI Publication
4. Machine Design, Sadhu Singh, Khanna Book Publishing Co.
5. Machine Design-R S Khurmi and J K Gupta, S Chand
6. J N Reddy "An Introduction to finite element method" Tata Mc Graw Hill 3rdedition
7. Machine Design Data Book, Sadhu Singh, Khanna Book Publishing Co.
8. S.S. Rao, "Finite Element Method In Engineering", PergamanPress
9. P Seshu, Finite Element Analysis, PHI publications,Delhi

Design data book: Design Data Handbook for Mechanical Engineering in SI and Metric Units – by K. Mahadevan, and K. Balaveera Reddy.

References:

1. Design of Machine Elements-M.F. Spott, PearsonEductaion
2. Machine Design-Maleev and Hartman, CBSPublishers.
3. Mechanical Engineering Design, 9e – Joseph E. Shigely, McGraw HillEducation.
4. Design of Machine Elements, GopePHI.
5. Finite Element Method with Applications in Engineering Y M Desai, PearsonPublication
6. Machine Design, Sadhu Singh, Khanna Book Publishing Co.
7. Introduction to Finite Element Analysis by TirupathiR.Chandrupatla& Ashok D Belegundu, PearsonPublication
8. Machine Design Data Book, Sadhu Singh, Khanna Book Publishing Co.
9. V.Ramamurti "Finite Element Method in Machine Design"Norosa PublishingHouse

THERMAL AND HYDRAULIC MACHINES

UNIT-I

Thermodynamic equilibrium, cyclic process, enthalpy, Zero, first and second laws of thermodynamics, Carnot cycle, concept of entropy, properties of steam, processes involving steam in closed and open systems, Enthalpy. Vapour Pressure Cycles:Rankine cycle, reheat cycle, Regenerative cycle

UNIT-II

Steam Turbine:Theoretical approach only of Classification, impulse and reaction turbines their velocity diagrams and related calculations, work done and efficiencies, re-heat factor, staging, bleeding and governing of turbines. Gas Turbine:Theoretical approach only of Classification, Brayton cycle, working principle of gas turbine, gas turbine cycle with intercooling, reheat and regeneration, stage and polytrophic efficiencies.

UNIT-III

Otto, Diesel .and Dual cycles, introduction to 2-stroke and 4-stroke SI and CI engines

UNIT-IV

Impact of Jet:Introduction to hydrodynamic thrust of jet on a fixed and moving surface (flat and curve). Hydraulic Turbines: Classification, heads and efficiencies, construction, working, work done and efficiency of impulse turbines.

UNIT-V

Centrifugal Pump:Classification, construction, working. Reciprocating Pump: Classification ,construction, working.

Text Books:

1. Onkar Singh “Applied Thermodynamics” New Age International,2006.

2. Steam & Gas Turbine by R.Yadav, CPHAllahabad

3. R.K.Rajput“ A Text Book of Hydraulic Machines” S. Chand & Co.,2008.

Reference Books:

4. P.L.Ballany “Thermal Engineering “ Khanna Publishers,2003

5. R.K.Bansal “A Text Book of Fluid Mechanics and Hydraulic Machines” Laxmi Publications,2006.

6. S.S Rattan, Fluid Mechanics and Hydraulic Machines, Khanna Book Publishing Co.

7. Gas Turbine, by V. Ganeshan, Tata McGraw Hill Publishers

APPLIED PROBABILITY AND STATISTICS

Unit I

Introduction to statistics: Statistical Thinking, Collecting data, Statistical Modeling Frame work, measure of central tendency and variance, Importance of Data summary and Display, Tabular and Graphical display.

Unit II

Discrete Random Variables and Probability distributions: Discrete Random variables, Probability distributions and Probability mass functions, Cumulative distribution functions, Mean and Variance of a discrete random variable, discrete uniform distribution, Binominal distribution, Hyper Geometric distribution, Poisson distribution, Applications.

Unit III

Continuous Random Variables and Probability Distributions: Continuous random variables, Probability distributions and probability density functions, cumulative distribution functions, Mean and Variance of a continuous random variable, uniform distribution, Normal distribution, Normal approximation to Binominal and Poisson distribution, Exponential distribution.

Unit IV

Testing of Hypothesis: estimation theory, Hypothesis testing, Inference on the mean of a population (variance known and unknown), Inference on the variance of a normal population, Inference on a population proportion, Testing for Goodness of Fit, Inference for a difference in Means, Variances known, Inference for a difference in means of two normal distributions, Variances unknown, Inference on the Variances of two normal populations, Inference on two population proportions.

Unit V

Simple Linear Regressions and Correlation: Simple Linear Regression, Properties of Least square Estimators and Estimation of variances, Transformations to a straight line, Correlation
Multiple linear regressions : Multiple linear regressions model, least square estimation of parameters, Correlation- types, correlation coefficients, properties of least square estimators and estimation of variance.

Text books:

1. George C Runger -Applied statistics and Probability for Engineers, John Wiley and Sons, ISBN-0-471-17027-5, 2ndEdn.
2. Richard I Levin, David S Rubin -Statistics for Management, Prentice Hall India, ISBN-81-203-0893-X, 6thEdn.

References:

1. William W Hines, Douglas C Montgomery -Probability and Statistics in Engineering, John Wiley and Sons, 2ndEdn.

2. Douglas C. Montgomery, George C. Runger, Norma F. Hubele, “Engineering Statistics”, John Wiley and Sons, 5th Edn, ISBN-13: 978-0470631478 ISBN-10: 0470631473 year:2010
3. Schaum's Outline of Statistics, 5th Edition (Schaum's Outlines) March 14, 2014, ISBN-13: 978-0071822527 ISBN-10:0071822526
4. Daniel, Terrell -Business Statistics for Management and Economics, Houghton Mifflin Company, ISBN-0-395-62835-0, 6thEdn.
5. Walpole & Mayer -Probability and Statistics, MacMillan Publishing Company,1989.

DESIG AND SIMULATIONLAB

1. Design & Modeling of Cotterjoint.
2. Design & Modeling of Knucklejoint
3. Design & Modeling of riveted joint applied to boilerjoints.
4. Study of a FEA package and modeling stress analysisof
   a. Bars of constant cross section area, tapered cross section area and steppedbar
   b. Trusses,
   c. Beams – Simply supported, cantilever, beams with UDL, beams with varying loadetc

5. **Mini Project:** Will take up problems from real life applications and optimize using modeling and analysissoftware

Each **s**tudent will be given a real life problem for the complete design of a subsystem/system

Using either manual calculation with the help of design handbook or through computer Programme or through software tools. This assignment should be done in groups, which will be submitted at the end of the semester.

Text Books:

1. Design of Machine Elements,V.B. Bhandari, Tata McGraw HillCo.
2. Design of Machine Elements, Sharma and Purohit,PHI.
3. Machine Design-R S Khurmi and J K Gupta, S Chand
4. J N Reddy "An Introduction to finite element method" Tata Mc Graw Hill 3rdedition
5. S.S. Rao, "Finite Element Method In Engineering", PergamanPress
6. P Seshu, Finite Element Analysis, PHI publications,Delhi

Design data book: Design Data Handbook for Mechanical Engineering in SI and Metric Units – by K. Mahadevan, and K. Balaveera Reddy

References:

1. Design of Machine Elements-M.F. Spott, PearsonEductaion
2. Machine Design-Maleev and Hartman, CBSPublishers.
3. Mechanical Engineering Design, 9e – Joseph E. Shigely, McGraw HillEducation.
4. Design of Machine Elements, GopePHI.
5. Finite Element Method with Applications in Engineering Y M Desai, PearsonPublication
6. Introduction to Finite Element Analysis by TirupathiR.Chandrupatla& Ashok D Belegundu, PearsonPublication
7. Machine Design, Sadhu Singh, Khanna Book Publishing Co.
8. V.Ramamurti "Finite Element Method in Machine Design"Norosa Publishing House
9. Machine Design Data Book, Sadhu Singh, Khanna Book Publishing Co.

THERMAL & HYDRAULIC MACHINE LAB

Experiments : Minimum 10 experiments out of following:

1. Study and working of Two stroke petrolEngine
2. Study and working of Four stroke petrolEngine
3. Study and working of two stroke DieselEngine
4. Study and working of four stroke DieselEngine.
5. Study of compounding of steamturbine
6. Study of Impulse & Reactionturbine
7. Impact of Jetexperiment.
8. Turbine experiment on Peltonwheel.
9. Turbine experiment on Francisturbine.
10. Turbine experiment on Kaplanturbine.
11. Experiment on Reciprocatingpump.
12. Experiment on centrifugalpump.

APPLIED PROBABILITY AND STATISTICS LAB

Software: MS Excel

1. Determining the measures of central tendency and dispersion of a givenprocess
2. Graphical/ Pictorial representation of data – Histogram, Ogive curve, Bar Chart, Column chart, Pie chart, Linechart
3. Interval estimation and hypothesis testing on single populationmean
4. Interval estimation and hypothesis testing on Two populationmeans
5. Interval estimation and hypothesis testing on single populationProportion
6. Interval estimation and hypothesis testing on Two populationProportions
7. Hypothesis testing using paired t –test
8. Construction of scatter plot and determination of kart Pearson's correlationcoefficient
9. Determination of Spearman's CorrelationCoefficient
10. Conduction of simple linear regressionanalysis
11. Testing the goodness of fit using Normaldistribution
12. Testing the goodness of fit using Binominal distribution calculationsthrough
13. Testing the goodness of fit using Poisson distribution MSExcel
14. Testing the goodness of fit using Uniformdistribution

Software: statistical software package

1. Determination of basic statistics and construction of histogram for a given quality characteristic
2. Determining the probability for discrete probability distributions – Hypergeometric, Binominal and Poissondistributions
3. Determining the probability for Continuous probability distributions – Normal and Exponentialdistributions
4. Interval estimation and Hypothesis testing on single populationmean.
5. Interval estimation and Hypothesis testing on two populationmeans
6. Interval estimation and Hypothesis testing on single populationProportion.
7. Interval estimation and Hypothesis testing on two populationProportions
8. Hypothesis testing using paired t –Test.
9. Construction of scatter plot and determining karl Pearson's correlationcoefficient.
10. Conduction of simple linear regressionanalysis.
11. Conduction of Multiple linear regressionanalysis.
12. Case study on engineering and management relatedproblems

Suggested software packages: MS Excel or LibreOffice Suite Calc (Using Spoken Tutorial MOOCs), SYSTAT / MINITAB 17 -Statistical Software Package etc

Department Elective Course –I

1. Mechtronics&Microprocessor.
2. ValueEngineering.
3. Maintenance Engineering &Management
4. Advance ManufacturingScience.
5. InventoryManagement

MECHATRONICS AND MICROPROCESSOR

UNIT I

Introduction to Mechatronic Systems: Measurement and control systems their elements and functions, Microprocessor basedcontrollers.

Review of Transducers and Sensors: Definition and classification of transducers. Definition and classification of sensors. Principle of working and applications of light sensors, proximity sensors and Hall effect sensors.

UNITII

Electrical Actuation Systems: Electrical systems, Mechanical switches, solid-state switches, solenoids, DC & AC motors, Stepper motors and their merits and demerits.

Signal Conditioning: Introduction to signal conditioning. The operational amplifier, Protection, Filtering, Wheatstone bridge, Digital signals Multiplexers, Data acquisition, Introduction to Digital system. Processing Pulse-modulation.

UNIT III

Introduction to Microprocessors: Evolution of Microprocessor, Organization of Microprocessors (Preliminary concepts), basic concepts of programming of microprocessors.

Review of concepts - Boolean algebra, Logic Gates and Gate Networks, Binary & Decimal number systems, memory representation of positive and negative integers, maximum and minimum integers. Conversion of real, numbers, floating point notation, representation of floating point numbers, accuracy and range in floating point representation, overflow and underflow, addition of floating point numbers, character representation.

UNIT IV

Logic Function: Data word representation. Basic elements of control systems 8085A processor architecture terminology such as CPU, memory and address, ALU, assembler data registers, Fetch cycle, write cycle, state, bus, interrupts. Micro Controllers. Difference between microprocessor and micro controllers. Requirements for control and their implementation in microcontrollers. Classification of micro controllers.

UNIT V

Organization & Programming of Microprocessors: Introduction to organization of INTEL 8085-Data and Address buses, Instruction set of 8085, programming the 8085, assembly language programming.

Central Processing Unit of Microprocessors: Introduction, timing and control unit basic concepts, Instruction and data flow, system timing, examples of INTEL 8085 and INTEL 4004 register organization.

REFERENCE BOOKS:

1. Mechatronics, W.Bolton, Longman, 2Ed, PearsonPublications,2007.
2. Microprocessor Architecture, Programming AndApplicationsWith 8085/8085A, R.S. Ganokar, WileyEastern.
3. Mechatronics and Microprocessors, K.P.Ramchandran, G.K.Vijayraghavan, M.S.Balasundran, Wiley, 1st Ed,2009
4. Mechatronics - Principles, Concepts and applications – Nitaigour and PremchandMahilik - Tata McGraw Hill-2003.
5. Mechatronics Principles & applications, Godfrey C. Onwubolu,Elsevier..
6. Introduction Mechatronics & Measurement systems,David.G.
7. Aliciatore& Michael. B. Bihistaned, Tata McGraw Hill,2000.

VALUE ENGINEERING

UNIT – 1

INTRODUCTION TO VALUE ANALYSIS: Definition of Value, Value Analysis, Value Engineering, Value management, Value Analysis versus Value Engineering, Value Analysis versus Traditional cost reduction techniques, uses, applications, advantages and limitations of Value analysis. Symptoms to apply value analysis, Coaching of Champion concept.
TYPE OF VALUES: Reasons for unnecessary cost of product, Peeling cost Onion concept, unsuspected areas responsible for higher cost, Value Analysis Zone, attractive features of value analysis. Meaning of Value, types of value & their effect in cost reduction. Value analysis procedure by simulation. Detailed case studies of simple products.

UNIT – II

FUNCTIONAL COST AND ITS EVALUATION: Meaning of Function and Functional cost, Rules for functional definition, Types of functions, primary and secondary functions using verb and noun, Function evaluation process, Methods of function evaluation. Evaluation of function by comparison, Evaluation of Interacting functions, Evaluation of function from available data, matrix technique, MISS technique, Numerical evaluation of functional relationships and casestudies.
PROBLEM SETTING & SOLVING SYSTEM: A problem solvable stated is half solved, Steps in problem setting system, Identification, Separation and Grouping of functions. Case studies.
PROBLEM SETTING & SOLVING SYSTEM: Goods system contains everything the task requires. Various steps in problem solving, casestudies.

UNIT – III

VALUE ENGINEERING JOB PLAN: Meaning and Importance of Value Engineering Job plan. Phases of job plan proposed by different value engineering experts, Information phase, Analysis phase, Creative phase, Judgement phase, Development planning phase, and case studies. Cost reduction programs, criteria for cost reduction program, Value analysis change proposal.
VALUE ENGINEERING TECHNIQUES: Result Accelerators or New Value Engineering Techniques, Listing, Role of techniques in Value Engineering, Details with Case examples for each of theTechniques.

UNIT-IV

ADVANCED VALUE ANALYSIS TECHNIQUES: Functional analysis system technique and case studies, Value Analysis of Management Practice (VAMP), steps involved in VAMP, application of VAMP to Government,
University, College, Hospitals, School Problems etc., (service type problems).

UNIT-V

TOTAL VALUE ENGINEERING: Concepts, need, methodology and benefits.
APPLICATION OF VALUE ANALYSIS: Application of Value analysis in the field of Accounting, Appearance Design, Cost reduction, Engineering, manufacturing, Management, Purchasing, Quality Control, Sales, marketing, Material Management Etc., Comparison of approach of Value analysis & other management techniques.

TEXT BOOKS:

1. Techniques of Value Analysis and Engineering – Lawrence D. Miles - McGraw Hill Book Company - 2ndEdn.
2. Value engineering for Cost Reduction and Product Improvement – M.S. Vittal - Systems Consultancy Services – Edn.1993
3. Value Management, Value Engineering and Cost Reduction – Edward D Heller - Addison Wesley Publishing Company-1971.

REFERENCE BOOKS:

1. Value Analysis for Better Management – Warren J Ridge – American Management Association - Edn1969
2. Getting More at Less Cost (The Value Engineering Way) – G.Jagannathan - Tata Mcgraw Hill Pub. Comp - Edn1995.
3. Value Engineering – Arther E Mudge - McGraw Hill Book Comp.-Edn1981.

MAINTENANCE ENGINEERING & MANAGEMENT
LTP 3-0-0

Unit-I

Introduction, operating life cycle, reliability, Failure data analysis, failure rate curve, hazard models, elements in series, parallel, mix, logic diagrams, improving reliability, redundancy-element, unit, standby, maintainability, availability, reliability and maintainability trade off.

Unit-II

Maintenance Strategies: Break down maintenance, planned maintenance, strategies, preventive maintenance, design out maintenance, planned lubrication, total productive maintenance, zero break down, preventive inspection of equipment used in emergency.

Unit-III

Replacement planning maintain or replace decision, replacement of items that deteriorate identical equipment, replacement of items that fail without deterioration individual, group replacement, replacement in anticipation of failure.

Unit-IV

Break down maintenance planning, assignment model, waiting time models expected waiting time, minimum cost service rate, PERT.

Unit-V

Maintenance Management, production maintenance system, objectives and functions, forms, policy, planning, organization, economics of maintenance, manpower planning, materials planning, spare parts planning and control, evaluation of maintenance management.

Books:

1. Management of systems – R.N. Nauhria& R.Prakash.
2. Operations Research –Wangner.
3. Engineering Management, S.C. Sharma & T.R. Banga, Khanna Book Publishing Co.

ADVANCE MACHINING PROCESSES

Unit I

Introduction: Limitations of Conventional machining processes, Need of advanced machining processes and its classification.
Mechanical Type Metal Removal Processes: Ultrasonic machining; Elements of the process; Tool design and economic considerations; Applications and limitations, Abrasive jet and Abrasive water jet machining principles; Mechanics of metal removal; Design of nozzles; applications, Abrasive finishing process, Magnetic abrasive finishing process

UNIT II

Thermal Type Advance Machining Processes: Classification, General principles and applications of Electro discharge, Plasma arc, Ion beam, Laser beam, Electron beam machining, Mechanics of metal removal in EDM, selection of EDM pulse generator dielectric, machining accuracy, surface finish and surface damage in EDM, Generation and control of electron beam for machining applications, advantages andlimitations

UNIT III

Chemical and Electro-chemical Type Metal Removal Processes: Principle, working advantages, disadvantages and applications of Electrochemical, Chemical machining, Economy aspects of ECM, Electro-chemical deburring and honing

UNITIV

Hybrid Unconventional Machining Processes: Introduction to ECDM, ECAM, Abrasive EDMetc.

UNIT V

Unconventional Forming processes: Principle, working and applications of High energy forming processes such as Explosive Forming, Electromagnetic forming, Electro Discharge forming, water hammer forming, explosive compaction etc.
Electronic-device Manufacturing: Brief description of Diffusion and Photo- Lithography process for electronic-device manufacturing

Books:

1. Advance Machining Processes V.K. Jain NewAge
2. Modern Machining Processes P.C. Pandey NewAge
3. Manufacturing Processes Degarmo-
4. Manufacturing Processes Kalpakjian Tata McGraw-HillInternational
5. Amitabh Gosh and A.K. Mallik, “Manufacturing Science”, Affiliated East-West Press Pvt. Ltd.,1985

INVENTORY MANAGEMENT

Unit I

Introduction: Definition & importance of materials management, MM at micro level & macro level, systems approach to MM, functions, interfaces & benefits of integrated systems approach.

Unit II

Purchasing: Principles, procedure, source & supplier selection, learning curve, negotiation, heding, forward buying, make or buy decision, legal aspects of buying, vendor vendee relations& vendor development.

Unit III

Inventory management: Definition, types, need & functions of inventory management, cost elements, Economic Ordering Quantity, ABC, FSN, VED analysis and spare parts management.
Deterministic models: Instantaneous replenishment model with & without shortages, production lot size model with & without shortages.

Unit IV

Deterministic/ Probabilistic models: With price breaks and quantity discounts, multi item deterministic model- restriction on floor space, total value and number of items, Probabilistic model, LIMIT technique.

Unit V
Stores Management and Operations: Storage System, Stores location and Layout. Development of Storing, Centralization and Decentralization of stores, Standardization and Variety reduction, codification, Materials accounting and materials audit.

Text books:

1. A.K. Chitale, R.C. Gupta - Materials Management A Supply Chain Perspective (Text and Cases), 3rd Edition PHI,2014.
2. P. Gopalakrishna& M. Sundaresan -Materials Management- an integrated approach, 1 st Edition, PHI,2012.
3. KantiSwaroop -Operation Research, S Chand &co.,2001.
4. A.K. Dutta -Materials Management, procedure, text and cases, prentice hall of India, new Delhi2006.

References:
1. Taha -Operation research, Mc Milan, ISBN0-02-418940-5,2001, 9th Edition,2012.
2. S.D.Sharma -Operation research, 4th Edition,2009.

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EVALUATION SCHEME

MECHANICAL AND INDUSTRIAL ENGINEERING

VI SEMESTER

Sl No.	Subject Code	Subject Name	Teaching Deptt.	L-T-P	Th/ Lab Marks	Sessional		Total	Credit
						Test	Assig/Att		
1	RAS601	INDUSTRIAL MANAGEMENT	Applied Science	3---0---0	70	20	10	100	3
2	RUC601/ RAS602	CYBER SECURITY/ SOCIOLOGY	Applied Science	3---0---0	70	20	10	100	3
3	RMI601	Work study and Ergonomics	Core Deptt.	3---0---0	70	20	10	100	3
4	RME604	Simulation Modelling & Analysis	Core Deptt.	3---1---0	70	20	10	100	4
5	RMI602	Machine Tool Design	Core Deptt.	3---0---0	70	20	10	100	3
6	RMI061,062, 063 RME065	DEPTTELECTIVE COURSE-2	Core Deptt.	3---1---0	70	20	10	100	4
7	RMI651	Work studyand Ergonomicslab	Core Deptt.	0---0---2	50		50	100	1
8	RME654A	Simulation Modelling & Analysis lab	Core Deptt.	0---0---2	50		50	100	1
9	RMI652	Machine Tool Design Lab	Core Deptt.	0---0---2	50		50	100	1
10	RMI-659	Seminar						100	1

Department Elective Course –2

RMI061: Human Factors in Engineering.
RMI062: Facilities Planning and Design.
RMI063: Software Engineering andManagement
RME065:Product Data Management

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WORK STUDY & ERGONOMICS

Unit I

Productivity: Definition of productivity, individual enterprises, task of management Productivity of materials, and, building, machine and power. Measurement of productivity, factors affecting the productivity, productivity improvement programs.
Work Study: Definition, objective and scope of work study. Human factor in work study Work study and management, work study and supervision, work study and worker.

Unit II

Introduction to Method Study : Definition, objective and scope of method study, activity recording and exam aids. Charts to record moments in shop operation – process charts, flow process charts, travel chart and multiple activity charts.(With simple problems)
Micro and Memo Motion Study : Charts to record moment at work place – principles of motion economy, classification of movements, two handed process chart, SIMO chart, and micro motion study. Development, definition and installation of the improved method.

Unit III

Introduction to Work Measurement: Definition, objective and benefit of work measurement. Work measurement techniques. Work sampling: need, confidence levels, sample size determinations, with simple problems.
Time Study: Time Study, Definition, time study equipment, selection of job, steps in time study. Breaking jobs into elements, recording information. Rating, Systems of rating.

Unit IV

Scales of rating, factors affecting rate of working, allowances and standard time determination. Predetermined motion time study – Method time measurement (MTM)
Wages and Incentives: introduction , definition, wage differentials ,methods of wage payment, Advantages ,disadvantages, Financial incentives, non-financial incentives.

Unit V

Ergonomics: Introduction, areas of study under ergonomics, man-machine system. Components of man-machine system and their functions –, study of development of stress in human body and their consequences. computer based ergonomics. Usability Engineering and Human Computer interface.
Design of man-machine system: Quantitative, qualitative representation and alphanumeric displays. types of control, layouts of panels and machines. Design of work places, influence of climate on human efficiency. Influence of noise, vibration and light on humanefficiency.

Text Books:

1. ILO -Introduction to work study, ISBN 13:9788120406025 Publisher: India Book House Pvt. Ltd, 4th Revised Edition,2008.

2. Ralph M Barnes -Motion and Time study, ISBN:13:978981426182 Publisher: John Wiley, 7th edition 2009**.**

References Books:

1. M S Sanders and E J McCormic -Human Factors in Engineering Design, ISBN: 13:9780070549012, Mc Graw Hill, 7th Edition,1992.

2. R.S.Bridger -Introduction to Ergonomics, ISBN:13:9780849373060, Publisher Taylor and Francis dated 20th Aug 2008, 3rdEdition.

SIMULATION MODELING AND ANALYSIS

Unit I

Introduction to Simulation: Simulation, advantages, Disadvantages, Areas of application, System environment, components of a system, Model of a system, types of models, steps in a simulation study.
Random Numbers: Properties, Generations methods, Tests for Random numbers - Frequency test, Runs test, Autocorrelation test, Gap test, Pokertest.

Unit II

Simulation Examples: Simulation of Queuing systems, Simulation of Inventory System, Monte Carlo simulation, General Principles, Concepts in discrete - events simulation, event scheduling / Time advance algorithm.

Unit III

Introduction to Probability distributions: Weibull, Triangular, Erlang and Gamma distributions and their applications (No analytical treatment)
Random Variate Generation:
Inverse Transform Technique- Exponential, Uniform, Weibull, Triangular distributions, Direct transformation for Normal and lognormal Distributions
Convolution Method – Erlang distribution Acceptance and Rejection technique – Poisson and Gammadistributions

Unit IV

Input Modeling: List of steps involved in input modeling – no analytical treatment Selecting input models without data, Multivariate and time series input models – Covariance and correlation, multivariate input models, time series input models .
Verification and Validation of Simulation Model: Model Building, Verification and validation, Verification of simulation models, Calibration and Validation of Models, Naylor and Finger's validation process.

Unit V

Output analysis for a single model: types of simulations, stochastic nature of output data, Output analysis of terminating simulations, Output analysis of steady state simulations
Optimization via simulation: What does "optimization via simulation" mean? Why is optimization so difficult? Basic GA and TS.

Text books:

1. Jerry Banks, John S Carson, II, Berry L Nelson, David M Nicol **-Discrete Event system Simulation**, III Edition, Pearson Education, Asia, ISBN - 81- 7808 – 505 -4.
2. NarsinghDeo**-Systems Simulation with Digital Computer**; PHI Publication (EEE), ISBN –0-87692-028-8

Reference:

1.Averill M Law, W David Kelton **-Simulation Modeling & Analysis,** McGraw Hill International Editions – Industrial Engineering series, ISBN – 0-07-100803-9.

MACHINE TOOL DESIGN

Unit-I

Introduction: Developments is machine tools, types of machine tools surface, profits and paths produced by machine tools. Features of construction and operations of basic machine tools e.g. lathe, drill, milling shapes and planers, grinding machine etc. General requirement of machine tool design. Machine tool design process. Tool wear, force Analysis.

Unit-II

Machine Tools Drives: Classification of machine tool drives, group Vs individual drives, election of electric motor, A brief review of the elements of mechanical transmission e.g. gear, belt and chain drives, slider-crank mechanism, cam mechanism, nut & Screw transmission, Devices for intermittent motion, reversing & differential mechanisms. Couplings and clutches Elements of hydraulic transmission system. e.g. pumps, cylinder, directional control valves, pressure valves etc. Fundamentals of Kinematics structure of machinetools.

Unit-III

Regulation of Speed and Feed rates: Laws of stepped regulation, selection of range ratio, standard progression ratio, selection of best possible structural diagram, speed chart, Design of feed box, Developing gearing diagrams. Stepless regulation of speed and feed in machine tool, speed and feed control.

Unit-IV

Design of Machine Tool Structure: Requirements and design criteria for machine tool structures, selection of material Basic design procedure for machine tool structures, design of bed, column and housing, Model technique in design. Design of guideways and power screws: Basic guideway profiles, Designing guideway for stiffness a wear resistance & hydrostatic and antifriction guideways. Design of sliding friction power Screws. Design of spindlier & spindle supports. 3 Layout of bearings, selection of bearings machinetools.

Unit-V

Dynamics of machine tools: General procedure for assessing the dynamic stability of cutting process, closed loop system, chatter in machine tools. Control Systems: Functions, requirements & types of machine tool controls, controls for speed & feed change. Automatic and manual Controls. Basics of numerical controls. Machine tool testing.

Books :

1. Machine Tools Design & Numerical Controls by N.K. Mehta,McGraw-Hill

2. Design of Machine Tools by S.K. Basu AlliedPublishers.

3. Principles of Machine Tools by Bhattacharya and Sen. New Central BookAgency.

4. Machine Tool Design Handbook by CMTI,McGraw-Hill.

WORK STUDY & ERGONOMICS LAB

List of Experiments

1. Construction of Outline Process Chart for simpleassembly
2. Recording the given activity using Flow ProcessChart
3. Recording the given activity using Multiple ActivityChart
4. Constructing the String Diagram for a shop-flooractivity
5. Construction of Two Handed Process Chart for pin board / Nut and Boltassembly.
6. Rating practice using walkingsimulator
7. Rating practice for dealing a deck ofcards
8. Determination of standard time/ rating of pin board assembly using centi minute stop watch.
9. Determination of standard time for simple operation using Timer ProSoftware
10. Measurement of parameters (heart beat rate, calorie consumption) using walking simulator
11. Measurement of parameters (heart beat rate, calorie consumption, revolutions per minute) usingergometer
12. Conduction of work sampling in office environment to determine standardtime.
13. Effect of noise, light and heat on human efficiency in workenvironment.

Text Books:

1. ILO **-Introduction to work study**, ISBN 13:9788120406025 Publisher: India Book House Pvt. Ltd, 4th Revised Edition,2008.
2. Ralph M Barnes **-Motion and Time study**, ISBN:13:978981426182 Publisher: John Wiley, 7th edition2009.

SIMULATION MODELING & ANALYSIS LAB

1. Introduction to Simulation Packages andselection.
2. Understanding the SimulationPackage
3. Identifying probability distributions for givendata
4. Building simulation models for manufacturing operations (Electronic assembly – With Basictemplates)
5. Building simulation models for manufacturing operations (Electronic assembly – With Commontemplates)
6. Building simulation models for manufacturing operations with transportSystem
7. Building simulation models for manufacturing operations withlayout
8. Building simulation models for manufacturing operations with layout and transport System
9. Building simulation Models for Banking service (Bank tellerproblem)
10. Building simulation Models for Mortgage applicationproblem
11. Building simulation Models for food processingproblem
12. Building simulation Models for Post officeanimation
13. Statistical Analysis of Simulation models (inputanalysis)
14. Statistical Analysis of Simulation models (outputanalysis)
15. Simulation model for foundryexercise.

Text books:

1. Jerry Banks, John S Carson, II, Berry L Nelson, David M Nicol -Discrete Event system Simulation, III Edition, Pearson Education, Asia, ISBN - 81- 7808 – 505 -4.

2. NarsinghDeo -Systems Simulation with Digital Computer; PHI Publication (EEE),ISBN – 0-87692-028-8

MACHINE TOOL DESIGN LAB

Minimum eight experiments out of the following:

1. Measurement and analysis of cutting forces in orthogonalturning.

2. Flank wear – time characteristics for single point cuttingtools.

3. (i) Checking the level of installation of a lathe in horizontal & vertical planes (ii) Checking the bed ways for straightness andparallelism.

4. Testing the main spindle of a lathe for axial movement and truerunning.

5. Process capability determination of a centerlathe.

6. Flatness checking of a surfaceplate.

7. A study of devices for intermittent motion used in machine tools e.g. ratchet gear & GenevaMechanism.

8. A study of Kinematics structure of lathe/millingmachine.

9. A study of the drives for reciprocation used in machinetools.

10. Development the speed chart and gearing diagram for a gassed headlathe.

11. A study of the cone pulley drive in center lathe and development of its ray diagram for the speedstructure.

12. Efficiency testing of lathe at variousparameters-values.

13. Accuracy analysis of finished cylindrical work-pieces produced on alathe.

14. Cutting (turning) with inclined placed tool (in toolfixture).

15. Turning with two simultaneously cutting tool (one from front on usual tool post and the other tool from back on tool-fixture oncarriage

Department Elective Course –2

1. Human Factors inEngineering.
2. Facilities Planning andDesign.
3. Product DataManagement
4. Software Engineering andManagement.
5. Quality Engineering InManufacturing.

HUMAN FACTORS IN ENGINEERING

Unit – 1

Introduction
Introduction to Human factors, History of Human factors, Human machine systems, Displaying information, coding of information, information processing, memory, decision making, age and information processing, mental workload.

Unit – II

Visual Displays: The process of seeing, Visual acuity types, quantitative visual displays, Dynamic quantitative displays, Qualitative visual displays, Representational displays

Unit – III

Auditory, Tactual & Olfactory Displays: The nature and measurement of sound, The anatomy of ear, Auditory displays, principles of auditory displays, Tactual display types , The Olfactory sense and displays.

Unit– IV

Human activities: Muscle physiology, Measure of physiological strain, physical work load, factors affecting, energy consumption, Strength and endurance, Biomechanics of human motion,
Function of controls, factors in control design, C/R ratio, Optimum C/R ratio, Principles of hand tool and device design.

Unit– V

Work space and arrangement: Anthropometry, use of anthropometric data, work spaces, design of work surfaces, science of seating, example of individual work place, human error, accidents and warnings
Applications: use of ergonomics in service sector and IT sector.
Introduction to BIS on Human factors.

Text Books:

1. M S Sanders and E J McCormick **-Human factors in Engineering & Design**, McGraw Hill, 7th Edition.
2. Wickens J. Lee, YD Liu, S GordanBeckere**-Introduction to Human factor in engineering**, Prentice Hall Inc, 2003.

References:

1. S Dalela and Sourabh**-Work Study and Ergonomics**, Standard publishers, 5th Revised & Enlarged Edition, 1999.

FACILITIES PLANNING AND DESIGN

Unit I

Plant Location and layout : Factors influencing plant location, location economics - problems. Objectives of plant layout, Principles of plant layout, types of plant layout, their merits and demerits. Line Balancing

Unit II

Material Handling : Objectives and principles of Material handling, Unit load concept, classification and types of material handling equipment, Modern material handling concepts and equipment, RFID. Muther's Systematic Layout Planning procedure – problems.

Unit III

Space Determination and Area Allocation : Factors for consideration in space planning, area allocation factors to be considered, Plot plan, Sequence demand Straight line and non directional methods – Analytical treatment**.** Determination of manpower and equipment requirement, use of travel chart for layout planning, analytical treatment.

Unit IV

Layout Evaluation: Methods of constructing the layout, efficiency indices.
Layout models: Single facility and multi facility location models, warehouse layout models, Warehouse design as per International standards. Conveyor and Storage models (discussion only)

Unit V

Computer Aided Layout: Introduction to CRAFT, COFAD, PLANET, CORELAP and ALDEP – Analytical treatment.

Layout for software and service organizations: An over view

Text Books:

1. James M Apple **-Plant Layout and Material handling**, 2nd Edition, John, Wiely and Sons.
2. Francies,R.L. and White,J.A**-Facility layout and Location**, Mc Graw Hill, 2ndedition.
3. Tompkins J A, White, Bozer and Tanchoco**-A Facilities planning**, John Wiley & Sons; 4th edition, 2010

References:

1. Muther Richard **-Practical Plant Layout**, Mc GrawHill-1955.
2. SundereshHeragu**-Facilities Design**, PWS Publishing Company, ISBN-0-534 -95183.
3. James M Moore **-Plant Layout Design**, Mac Millan Co.1962 LCCCN61-5204.

PRODUCT DATA MANAGEMENT

UNIT I

Introduction: Introduction to PDM-present market constraints need for collaboration-Internet and developments in server-client computing, Collabrative product commerce

UNIT II

Components of PDM: Components of a typical PDM set-up hardware and software-document management creation and viewing of documents -creating parts-version control of parts and documents –case studies.

UNIT III

Configuration Management: Base lines-product structure configuration management -case studies.
Projects and Roles: Creation of projects and roles -life cycle of a product- life cycle management –automating information flow -work flows-Creation of work flow templates - life cycle -work flow integration -case studies.

UNIT IV

Change Management: Change issue -change request-change investigation- change proposal-change activity-case studies.

UNIT V

Generic Products and Variants: Products configuration comparison between sales configuration mild products generic-generic product modeling in configuration mode1er-use of order generator for variant creation –registering ofvarian1s in product register-case studies.

REFERENCE BOOKS:
1. **Computer Integrated Design and Manufacturing** - David Bed worth. Mark Henderson &.Philips
Wolfe - McGraw Hill Inc. - 1991.
2. **Visual Modeling with Rational Rose and UML** - Terry Quatrain - Addison Wesley - 1998.
3. **Wind-chill -** RS.O Reference manuals -2000.

SOFTWARE ENGINEERING & MANAGEMENT

Unit I

The Product and the Process: The product - Evolving role of Software, Characteristics, Components, and Applications. The Process - Software process, Models - Linear, sequential, Prototype, RAD, Process Technology, Software Development Life cycle.

Unit II

Software Project Management Concepts: The Management Specification, People, Problem,Process.
Software Project Planning : Objectives, Scope, Resources, Project estimation, Decomposition Techniques, Empirical Estimation Models, Make-buy decision, Automated estimation tools.

Unit III

Risk Management : Reactive v/s Proactive Risk Strategies, Software Risks, Risk identification, Risk projection, Monitoring.
Software Project Scheduling and Tracking: Basic concepts, defining a task set selection, Defining Scheduling, Project Plan.

Unit IV

Software Quality Assurance/Quality Control: Quality assurance concept, Cost impact of software defects, Technical review, statistical Quality assurance, software reliability, ISO 9000 Quality standards.
System Analysis Concept and Principles: Requirement analysis, Principles, software prototyping, specifications, Data Modeling, Functional Modeling and Information Flow, Structured Analysis, Data Dictionary.

Unit V

System Design Concepts and Principles: Design Process Concept, Modular design, Documentation Design Methods, Data Design, Interface Design, Procedural Design, Design for Real Time Systems, Considerations in Real Time systems, Analysis and Simulation of Real Time Systems.

Text Book:

1. Roger S. Pressman **-Software Engineering**: A Practitioner's Approach, Software Engg. Series, McGraw-Hill International Editions , 4th Ed.
2. Nasib Singh Gill, Software Engineering, Khanna Publishing House

Reference:

1. Somerville **-Software Engineering**, Adison Wesley publications, 8th edition

QUALITY ENGINEERING IN MANUFACTURING

UNIT I

Quality Value and Engineering: An overall quality system, quality engineering in production design, quality engineering in design of production processes. Loss Function and Quality Level: Derivation and use of quadratile loss function, economic consequences of tightening tolerances as a means to improve quality, evaluations and types tolerances.(N-type,S-type and L-type)

UNIT II

Tolerance Design and Tolerancing: Functional limits, tolerance design for N-type. L-type and S-type characteristics, tolerance allocation fbr multiple components. Parameter and Tolerance Design: Introduction to parameter design, signal to noise ratios, Parameter design strategy, some of the case studies on parameter and tolerance designs.

UNIT III

Analysis of Variance (ANOVA): NO-way ANOVA, One-way ANOVA, Two-way ANOVA, Critique of Ftest, ANOVA for four level factors, multiple level factors.

UNIT IV

Orthogonal Arrays: Typical test strategies, better test strategies, efficient test strategies, steps in designing, conducting and analyzing an experiment. Interpolation of Experimental Results: Interpretation methods, percent contributor, estimating the mean.

UNIT V

IS[)-9000 Quality System, BDRE, 6.-sigma, Bench making, Quality circles Brain Storming — Fishbone diagram — problem analysis.

Text Book:

1. Taguchi Techniques for Quality Engineering / Phillip J. Ross / McGraw Hill, Intl. II Edition, 1995.

Reference Books:

1. Quality Engineering in Production systems I G. Taguchi, A. Elsayed et al / Mc.Graw Hill Intl. Edition,1989.

2. Taguchi Methods explained: Practical steps to Robust Design / Papan P. Bagchi I Prentice Hall md. Pvt. Ltd., New Delhi.

3. Design of Experiments using the Taguchi Approach/Ranjit K. Roy, John wiley& sons. Inc.2001.